Ch. 2: specific heat of solids

monoatomic ideal gas $C_V = \frac{3}{2} k_B$ per atom.

Experiments: in solids $C_V = 3 k_B$ per atom. (Dulong-Petit)

1896 Boltzmann $\rightarrow$ atoms interact harmonically

$$\Rightarrow H = \sum \frac{\vec{p}_i^{\,2}}{2m} + \frac{1}{2} k (\vec{x}_i - \vec{x}_j)^2$$

$\Rightarrow$ 6 degrees of freedom per atom

$$\Rightarrow C_V = 6 \cdot \frac{1}{2} k_B = 3 k_B .$$

1907 Einstein: Dulong-Petit does not work at low T. why?

Assume that every atom sits in harmonic well (created by other atoms)

$$H_{atom} = \frac{\vec{p}^{\,2}}{2m} + \frac{1}{2} m \omega^2 x^2$$

$\hookrightarrow$ Einstein frequency

$$\Rightarrow E_n = \hbar\omega \left(n + \frac{1}{2}\right)$$

$$Z_1 = \sum_{n \geq 0} e^{-\beta \hbar \omega (n + \frac{1}{2})} = e^{-\beta \frac{\hbar\omega}{2}} \sum_{n=0}^{\infty} \left(e^{-\beta\hbar\omega}\right)^n =$$

$$= \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}} = \frac{1}{2 \sinh\left(\frac{\beta\hbar\omega}{2}\right)}$$

Average $\langle E \rangle = -\frac{\partial}{\partial \beta} \ln Z = \frac{\partial}{\partial \beta} \ln\left(2 \sinh \frac{\beta\omega\hbar}{2}\right)$

$$= \frac{\cosh \frac{\beta\hbar\omega}{2}}{\sinh \frac{\beta\hbar\omega}{2}} \frac{\hbar\omega}{2} = \frac{\hbar\omega}{2} \frac{e^{\beta\hbar\omega} + 1}{e^{\beta\hbar\omega} - 1} =$$

$$= \frac{1}{2} \hbar\omega \left(\frac{2}{e^{\beta\hbar\omega} - 1} + 1\right) = \hbar\omega \left(\underbrace{\boxed{\frac{1}{e^{\beta\hbar\omega} - 1}}}_{n_B(\hbar\omega)} + \frac{1}{2}\right)$$

Bose occupation factor $n_B(x) = \frac{1}{e^{\beta x} - 1}$

Thus $\langle E \rangle = \hbar\omega \left(n_B(\hbar\omega) + \frac{1}{2} \right)$

cf. $E_n = \hbar\omega \left(n + \frac{1}{2} \right)$

$\langle E \rangle$ describes oscillator excited to n_B-th level.

OR: oscillator with n_B bosons, each of which comes with energy $\hbar\omega$.

$$C = \frac{\partial \langle E \rangle}{\partial T} \qquad \frac{\partial}{\partial T} = \frac{\partial}{\partial \beta} \underbrace{\frac{\partial \beta}{\partial T}}_{-\frac{1}{k_B T^2} = -k_B \beta^2} = -k_B \beta^2 \frac{\partial}{\partial \beta}$$

$$C = -k_B \beta^2 \frac{\partial}{\partial \beta} \left(\frac{\hbar\omega}{e^{\beta\hbar\omega} - 1} + \frac{\hbar\omega}{2} \right) = k_B \beta^2 (\hbar\omega)^2 \frac{e^{\beta\hbar\omega}}{\left(e^{\beta\hbar\omega} - 1\right)^2}$$

Check the high-T limit: $T \to \infty \Rightarrow \beta \to 0$

$$C \xrightarrow[\beta \to 0]{} k_B (\beta\hbar\omega)^2 \frac{1 + \beta\hbar\omega + \ldots}{\left(1 + \beta\hbar\omega + \ldots - 1\right)^2} = k_B$$

(Dulong-Petit!)

But low-T limit: $T \to 0 \Rightarrow \beta \to \infty$

$$C \xrightarrow[\beta \to \infty]{} k_B (\beta\hbar\omega)^2 \frac{e^{\beta\hbar\omega}}{e^{2\beta\hbar\omega}} = k_B (\beta\hbar\omega)^2 e^{-\beta\hbar\omega}$$

exponential suppression for $\beta\hbar\omega \gg 1 \Rightarrow \underline{\hbar\omega \gg k_B T}$

For $d = 3$ oscillator $E_{n_x, n_y, n_z} = \hbar\omega \left(n_x + \frac{1}{2} + n_y + \frac{1}{2} + n_z + \frac{1}{2} \right)$

$$Z_3 = \sum_{n_x=0}^{\infty} \sum_{n_y=0}^{\infty} \sum_{n_z=0}^{\infty} e^{-\beta\hbar\omega (n_x + n_y + n_z + 3/2)}$$

$$= (Z_1)^3 \Rightarrow \langle E_3 \rangle = 3 \langle E_1 \rangle$$

$$\Rightarrow \quad C = 3k_B (\beta\hbar\omega)^2 \frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega}-1)^2}.$$

$$C \longrightarrow \begin{cases} 3k_B & \text{at high } T \gg \frac{\hbar\omega}{k_B} \\ 3k_B(\hbar\omega\beta)^2 e^{-\beta\hbar\omega} & \text{at low } T \ll \frac{\hbar\omega}{k_B} \end{cases}$$

Experiments on non-magnetic insulators suggest $C \sim T^3$ at low T.

1912 Debye — the oscillations are sound waves

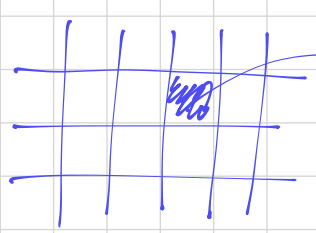
therefore frequency depends on wave number

$$\omega(\vec{k}) = \underbrace{v}_{\text{sound velocity}} k \quad , \quad k = |\vec{k}|.$$

Think of oscillator with ω_k for every possible $\vec{k}$

$$\Rightarrow \quad \langle E \rangle = 3 \sum_{\vec{k}} \hbar\omega(\vec{k}) \left(\frac{1}{e^{\beta\hbar\omega(\vec{k})}-1} + \frac{1}{2} \right)$$

$$\vec{k} = \frac{2\pi}{L}(n_1, n_2, n_3)$$



$\left(\frac{2\pi}{L}\right)^2$ in $d=2$

$$\sum_{k_x} \longrightarrow \int \frac{dk_x}{\Delta k_x} = \frac{L}{2\pi} \int dk_x$$

$$\Delta k_x = \frac{2\pi}{L}$$

$$\sum_{\vec{k}} \longrightarrow \frac{L^3}{(2\pi)^3} \int \underbrace{dk_x\, dk_y\, dk_z}_{d\vec{k}}$$

Since $\omega(\vec{k})$ depends on magnitude $k = |\vec{k}|$ only

we can use spherical symmetry $d\vec{k} \rightarrow 4\pi k^2 dk$

$$\langle E \rangle = 3 k_B \frac{L^3}{(2\pi)^3} \int_0^\infty dk \; 4\pi k^2 \; \hbar\omega(\vec{k}) \left(n_B(\hbar\omega(\vec{k})) + \frac{1}{2} \right)$$

$$k = \frac{\omega}{v}$$

$$= 3 k_B \frac{4\pi L^3}{(2\pi)^3 v^3} \int_0^\infty d\omega \; \omega^2 \; \hbar\omega \left(\frac{1}{e^{\beta\hbar\omega}-1} + \frac{1}{2} \right)$$

$$= \int_0^\infty d\omega \; g(\omega) \; \hbar\omega \left(\frac{1}{e^{\beta\hbar\omega}-1} + \frac{1}{2} \right)$$

where density of states $g(\omega) = L^3 \frac{12\pi\omega^2}{(2\pi)^3 v^3} = N \frac{9\omega^2}{\omega_d^3}$

where $N = n L^3$ is the number of atoms (n = density)

$\omega_d = 6\pi^2 n v^3$ Debye frequency.

The energy can be estimated by scaling: let $\beta\hbar\omega = x$

$$\omega = \frac{x}{\beta\hbar} = \frac{k_B T}{\hbar} x$$

$$\langle E \rangle = \frac{9N}{\omega_d^3} \int_0^\infty dx \; x^3 \left(\frac{1}{e^x - 1} + \frac{1}{2} \right) \left(\frac{k_B T}{\hbar} \right)^3 (k_B T)$$

$$= \frac{9N (k_B T)^4}{(\hbar\omega_d)^3} \int_0^\infty dx \; x^3 \left(\frac{1}{e^x - 1} + \frac{1}{2} \right)$$

$$C = \frac{\partial}{\partial T} \langle E \rangle \sim T^3. \quad \text{(but at all } T\text{!)}$$

But how to reproduce the high-T limit of Dulong-Petit: $C \to 3k_B$?

<u>Debye</u> made a very good guess: at most there are 3 oscillatory degree of freedom per atom

$\Rightarrow$ there must be cut-off frequency ω_{cutoff} such that

$$3N = \int_0^{\omega_{cutoff}} d\omega\, g(\omega) = 9N \int_0^{\omega_{cutoff}} d\omega \frac{\omega^2}{\omega_d^3} = 3N \frac{\omega_{cutoff}^3}{\omega_d^3}$$

$$\Rightarrow \omega_{cutoff} = \omega_d = \left(6\pi^2 n v^3\right)^{1/3} \qquad n = \frac{N}{L^3} \text{ density of atoms}$$

Now $$\langle E \rangle = \int_0^{\omega_{cut\,off}} d\omega\, g(\omega) \left(n_B(\omega) + \frac{1}{2}\right) \cdot \hbar\omega =$$

$$= \frac{9N\hbar}{\omega_d^3} \int_0^{\omega_{cutoff}} d\omega \frac{\omega^3}{e^{\beta\hbar\omega} - 1} + T\text{-independent constant}$$

$$\beta\hbar\omega = x, \quad \omega = \frac{x}{\beta\hbar} = \frac{k_B T}{\hbar} x$$

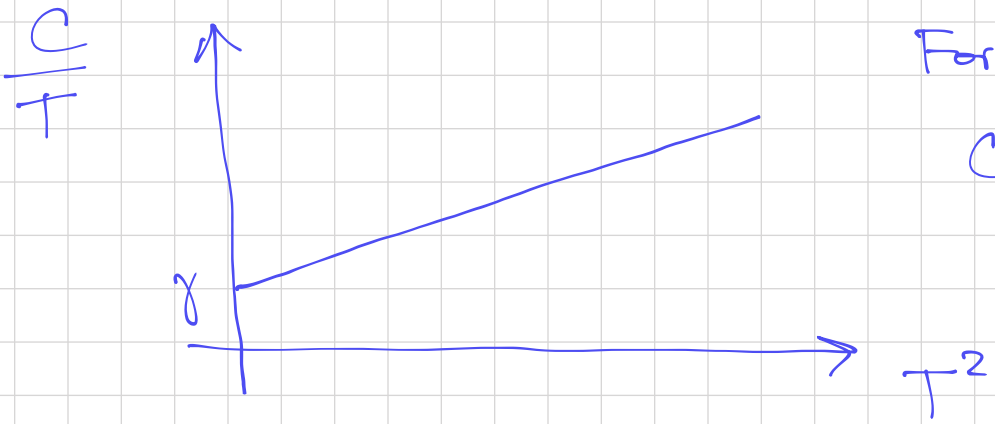
$$\langle E \rangle = \frac{9N\hbar}{\omega_d^3} \left(\frac{k_B T}{\hbar}\right)^4 \int_0^{\beta\hbar\omega_{cutoff}} dx \frac{x^3}{e^x - 1}$$

$$\beta\hbar\omega_{cutoff} = \frac{\hbar\omega_{cutoff}}{k_B T} \Rightarrow \begin{cases} \infty, & \text{low } T \\ \ll 1 & \text{for high } T \end{cases}$$

At high T $x \in \left(0, \frac{\hbar\omega_{cutoff}}{k_B T} \ll 1\right) \Rightarrow$ we can expand $n_B =$

$$= \frac{1}{e^x - 1} \simeq \frac{1}{1 + x + \ldots - 1} = \frac{1}{x}$$

$$\Rightarrow \langle E \rangle = \frac{9N\hbar}{\omega_d^3} \left(\frac{k_B T}{\hbar}\right)^4 \underbrace{\int_0^{\beta\hbar\omega_{cutoff}} dx\, x^2}_{\frac{1}{3}\left(\frac{\hbar\omega_d}{k_B T}\right)^3} = 3N k_B T \Rightarrow \frac{C}{N} = 3k_B.$$



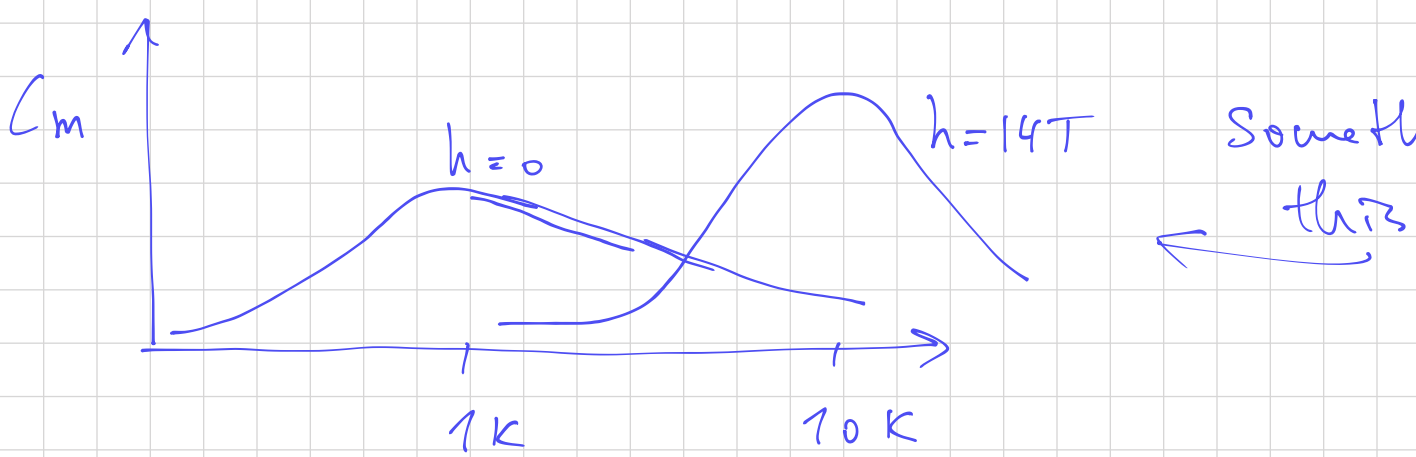
For conductors

$$C = \underbrace{\gamma T}_{\text{electrons}} + \underbrace{\alpha T^3}_{\text{phonons}}$$

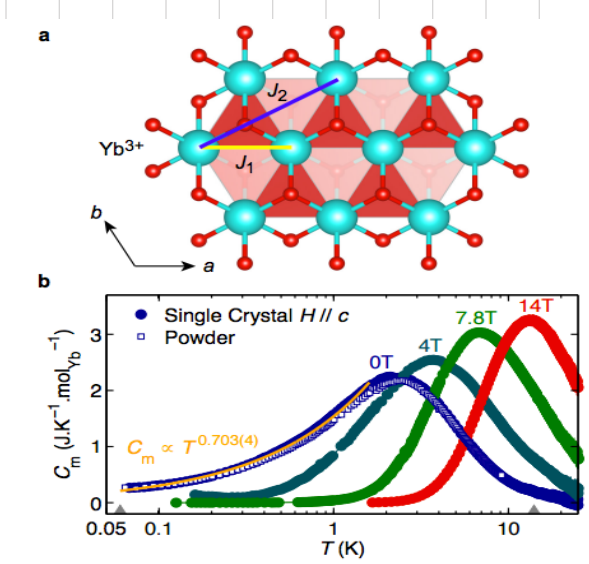
Interesting recent example: magnetic insulator $YbMgGaO_4$

<u>Magnetic</u> contribution to specific heat, C_m, depends on magnetic field h in the interval 0 to 14 Tesla, and at temperature 50 mK to 10 K (Fig. 1b in arxiv:1607.03231

$$C_m \sim T^{0.7} \text{ for } T \leq 1\,K.$$



Something like this



This is the data from Paddison et al.